

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

CONSTRAINT-BASED PROBLEM SOLVING

Overview

This report summarizes the approach, method, and results obtained for the three problems in the Constraint-Based Problem Solving assessment. The solutions were developed using appropriate Artificial Intelligence techniques including Backtracking Search, Breadth-First Search (BFS), and Uniform Cost Search (UCS). The corresponding Python implementations verify the correctness of the obtained solutions.

1. Hospital Doctor Shift Scheduling

Method

Backtracking Search was used to assign doctors to available shifts while satisfying all scheduling constraints. The algorithm systematically explores possible assignments and backtracks whenever a constraint violation occurs.

Results

Metric	Value
Search Strategy	Backtracking Search
Number of Doctors	3
Number of Shifts	3
Constraints Satisfied	5
Final Schedule	D1→Morning, D2→Afternoon, D3→Night

Valid Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

2. Robot Navigation in a Grid Environment

Method

Breadth-First Search (BFS) was used to explore the 5×5 grid environment. BFS guarantees the shortest path in an unweighted graph by expanding nodes level by level until the goal state is reached.

Results

Metric	Value
Search Strategy	Breadth-First Search
Grid Size	5 × 5
Start Node	(0,0)
Goal Node	(4,4)
Path Cost	8
Shortest Path Found	Yes

Optimal Path

(0,0) → (0,1) → (0,2) → (0,3) → (0,4) → (1,4) → (2,4) → (3,4) → (4,4)

3. Autonomous Rescue Robot

Method

Uniform Cost Search (UCS) was selected because the environment contains varying path costs. The robot evaluates movement costs and additional risk-zone penalties to determine the least-cost route to the survivor.

Results

Metric	Value
Search Strategy	Uniform Cost Search
Environment Type	Partially Observable
Risk Zones Considered	Yes
Survivor Located	Yes
Goal Reached	Yes
Minimum Cost Path	Obtained

AI Concepts Applied

Concept	Application
Intelligent Agent	Rescue Robot
Percepts	Adjacent Cell Observations
Actions	Move Up, Down, Left, Right
Rationality	Chooses Least-Cost Path
Environment	Dynamic and Partially Observable

Conclusion

The three problems were successfully solved using suitable Artificial Intelligence techniques. Backtracking Search efficiently handled the scheduling problem, Breadth-First Search identified the shortest navigation path, and Uniform Cost Search enabled optimal rescue planning in a cost-sensitive environment. The obtained results satisfy all given constraints and demonstrate the effectiveness of AI search strategies in solving real-world problems.
